Mental-health emergency visits, community follow-up and acute psychiatric beds across Chile's Health Services, 2024

C1 · Business Intelligence (IIB423T-1) · Universidad del Desarrollo

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Problem and decision

Main question. In 2024, how did the 29 Health Services differ in mental-health emergency visits, community follow-up and acute psychiatric bed use, and which area deserves a closer look in each?

Proposed user. Mental-health unit of a Health Service; MINSAL network division.

Decision. Review first: (a) community follow-up, (b) acute bed capacity, (c) organisation of emergency and referral care.

Scope. Public network, 2024, visits and bed-days, **not people**.

From W1 to C1

W1	C1, after feedback
Stress ratio: visits / bed-days	Visits, bed-days and occupancy shown separately
Law 21.331 as a mandate being breached	Law and ENS 2030 as context only
"15% target", "2.4% budget"	Removed: not in the sources
Emergency vs beds	+ community follow-up (REM A06)
640 vs 609 facilities	Resolved: 609 legacy + 31 modern codes

Data: three sources, one territorial code

Pathway stage	Source	One row =	Observed in 2024
Community care	REM A06 section A.1	facility × month × professional	Yes: 5.3 M follow-up visits
Emergency visit	SADU causes 36, 35	facility × day × cause	Yes: 305,431 + 9,753 visits
Decision to hospitalise	SADU cause 42	facility × day	Yes: 27,306, hospital EDs only
Waiting for a bed	SADU causes 27, 28	facility × day	**No: zero on every facility-day**
Acute psychiatric stay	REM20, 4 acute areas	hospital × area × month	Yes: 1,130 beds, 89.3% occupied

No patient identifier: sources are aggregated to Health Service × 2024 and compared side by side, not joined row by row.

Facility join SADU → registry: 640/640 facilities, N:1, no duplication. Same 29 Health Service codes in all sources; names differ (e.g. "Iquique" = "Tarapacá").

Quality checks that changed the analysis

- **Cause hierarchy.** Cause 36 = 37+38+39+40+41 on all 224,330 facility-days. 36 is the mental-health total and includes suicidal ideation. W1's 35+37+39+40 mixed levels.
- **Cause 42** is not a visit: 25 = 7+8+22+23+32+33+42 on every facility-day, so 42 = ED hospitalisations for mental disorders.
- **Recorded zeros.** "Patients waiting for a bed" is zero everywhere, so waiting cannot be measured.
- **Acute beds.** Four acute areas (mean stay 17–26 days); long-stay and forensic areas excluded. W1 had missed psychiatric intensive care.
- **Cleaning.** 24 duplicate rows removed, 28 codes stripped, 1 facility's blank Health Service filled from the registry. Occupancy recomputed from bed-days matches the published index.

National picture, 2024

1.57%

of 19.4 M emergency visits are mental-health visits

17.3

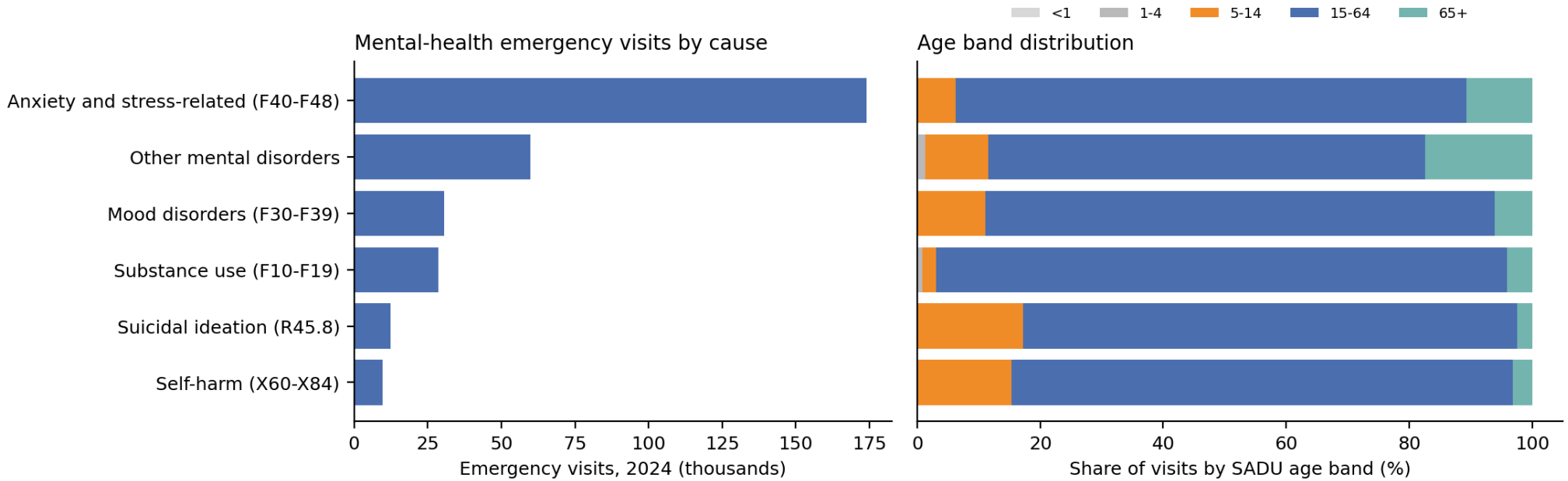
ED hospitalisations per 100 mental-health visits at hospital EDs

89.3%

acute psychiatric occupancy (1,130 beds on average)

17.4

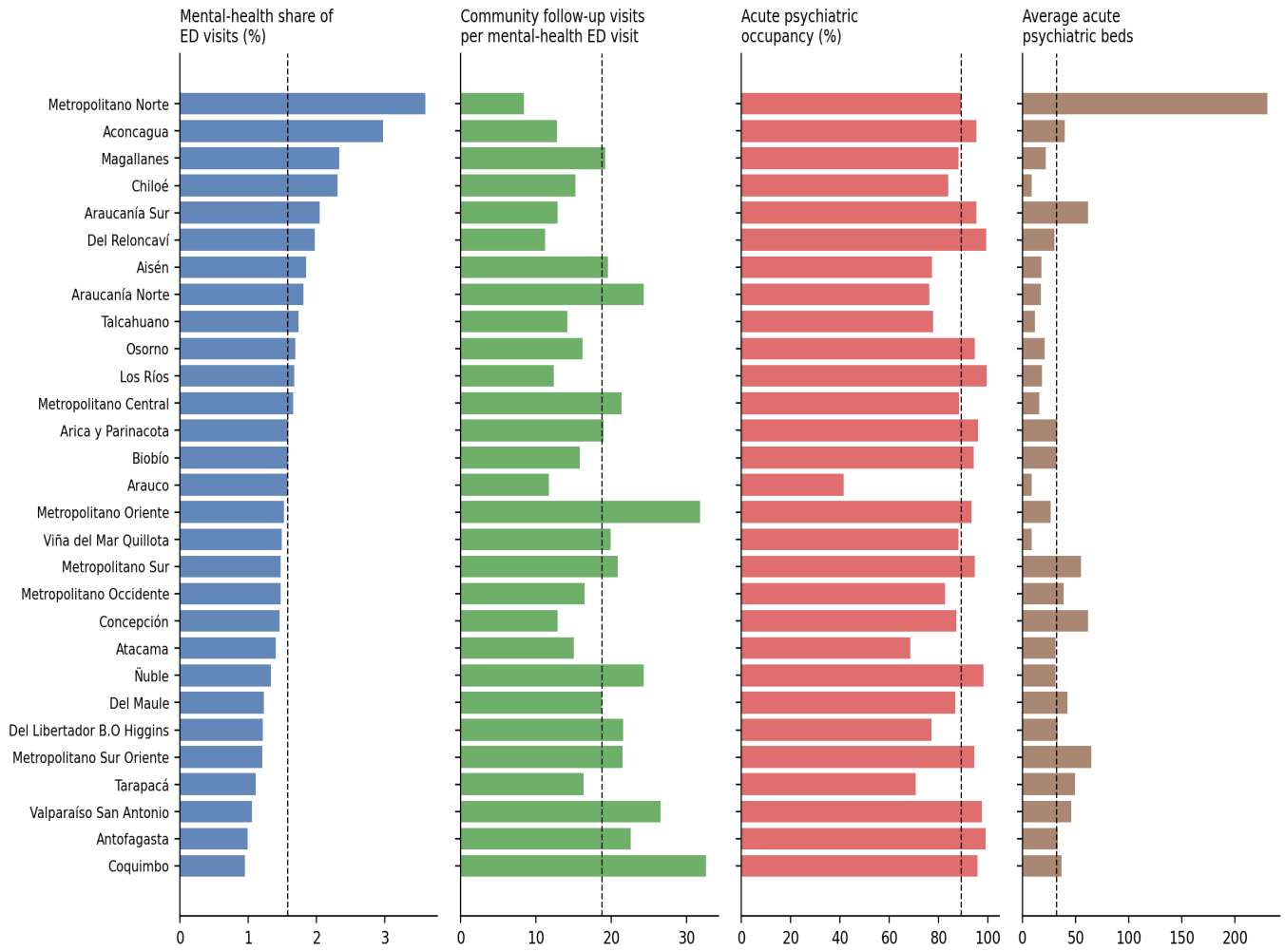
community follow-up visits per mental-health ED visit



Source: DEIS Atenciones de Urgencia 2024. Visits, not people. 15-64 is the narrowest adult band SADU publishes.

Anxiety/stress-related: 57% of visits. 5–14 band: 17% of suicidal ideation, 15% of self-harm. 15–64 includes ages 15–17.

Health Service profiles



Dashed line: median of the 29 Health Services. Sources: DEIS SADU 2024, REM20 2024 (acute psychiatric areas), REM A06 section A.1 2024.

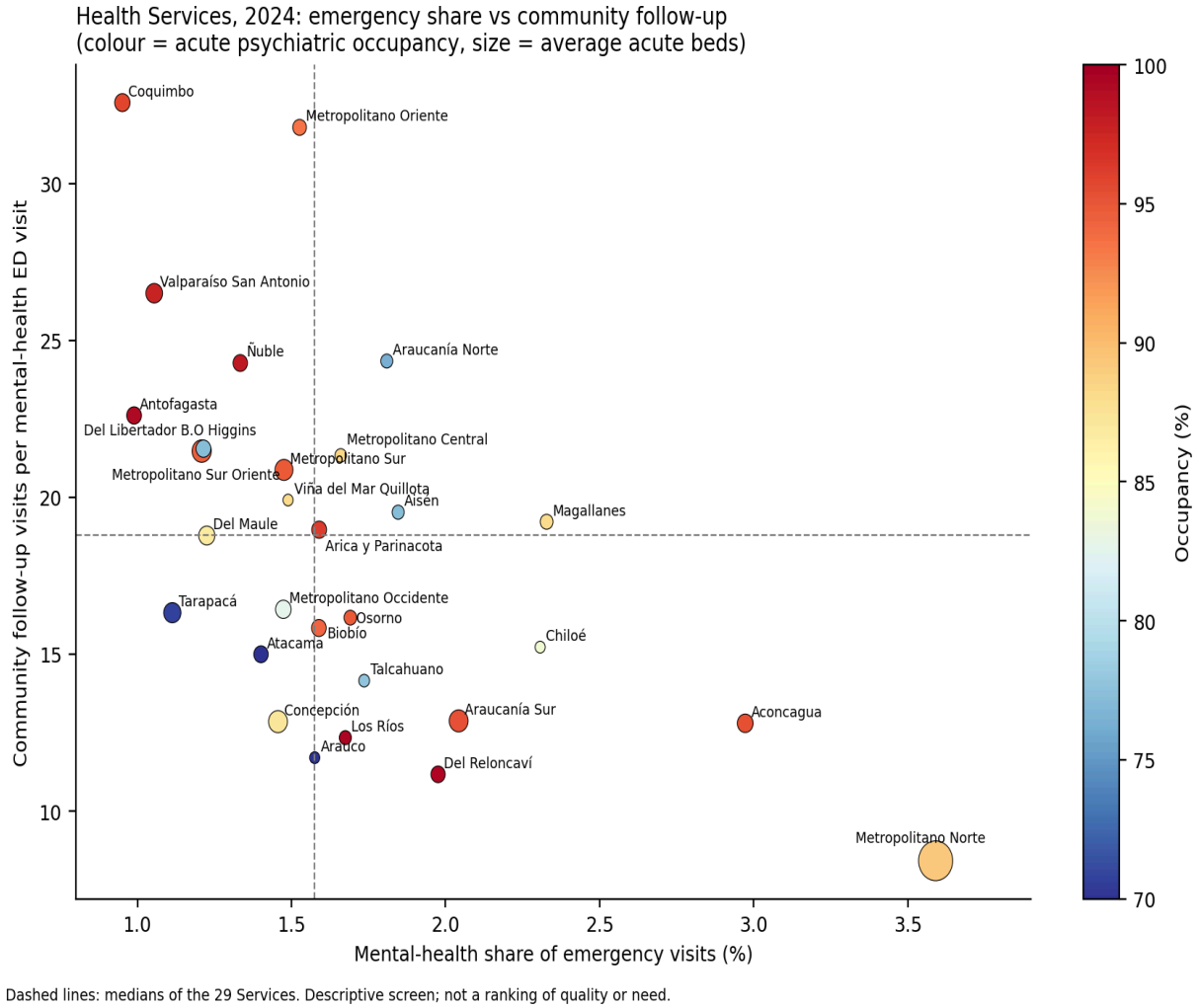
Follow-up per ED visit: 8.4 to 32.6 (median 18.8).

Occupancy: 41.5% to 99.6%.

Concentration. The Horwitz Institute has 55% of Metropolitano Norte's mental-health visits; Philippe Pinel has 61% of Aconcagua's. Without them the shares fall from 3.6% to 1.7% and from 3.0% to 1.2%, so the high shares reflect referral centres, not local need.

Check before reading as care: Araucanía Norte 0.7 ED hospitalisations per 100 (median 21.1); Arauco 41.5% occupancy.

Screen: where to look first



Flags: emergency share above the median (without psychiatric EDs), follow-up below the median, occupancy ≥ 90%.

Flags	Services
3	Biobío, Araucanía Sur, Los Ríos, Osorno, Del Reloncaví
2	Arica y Parinacota, Aconcagua, Metropolitano Norte, Talcahuano, Arauco, Chiloé
0	Viña del Mar Quillota, O'Higgins, Del Maule

Occupancy flag: 20 Services at 85%, 14 at 90%, 9 at 95%. A descriptive screen, not a ranking of quality or need.

What the evidence supports, and what it does not

Observed. Several-fold differences in follow-up and occupancy. Five Services combine a high emergency share, low follow-up and full beds.

Interpretation. For those five, review community follow-up first, then bed management.

Hypotheses. Low follow-up leads to more crises reaching the ED; long stays keep occupancy high.

Alternatives. Recording practice; referrals between Services; private sector; population size and age.

Cannot establish: need, quality, waiting time, causality, or whether a person was followed up after an ED visit.

W1 reconciled. Del Maule 1.83 reproduces, but is 1.17 with acute beds. "18 Services without child beds" is reversed: 18 have child short-stay beds; 7 report no acute child area (not reported ≠ no service).

Needed next: linked ED-admission records, a population denominator, 2021–2023 data, confirmation of anomalies.

Continuity: machine learning and dashboard

ML: clustering Health Service profiles

- Unit: Health Service × year, 2021–2024 (116 rows; data already available).
- Inputs: the five indicators, primary-care share of follow-up, child share of acute beds.
- No labels; clusters described. A Service compares itself with similar peers.
- For review: an occupancy ≥ 95% month, predicted from previous months only (no same-month leakage).

Dashboard

- Users: Health Service mental-health unit; MINSAL network division.
- Views: screen scatter; Service profile by month; data-quality panel.
- Filters: year, month, cause, age band, facility type, with or without psychiatric EDs.
- Waiting left out until a source records it.

Process record

Feedback response. 9 of 10 items accepted, 1 adapted: the DEIS 2024 establishment list could not be downloaded from our environment, so the 2026 registry was used and the gap stated.

AI use. W1: Gemini (scaffolding, translation). C1: Claude (extraction, notebook, drafting). Unsupported W1 claims removed after source checks. Every number comes from the executed notebook, and Service totals reconcile with national totals.

Contributions

Member	Contribution
Benjamín Buzeta	[TO COMPLETE]
Oscar Pichuman Millape	[TO COMPLETE]